



MTH101, Basic Linear Algebra

Monsoon 2025

04/09/2025, 2:00PM-3:00PM.

Tutorial-1

- (1) Draw the solution sets of the following three equations¹ in $\mathbb{R}^2$.
 - (a) $2X+Y=2$
 - (b) $Y=0$
 - (c) $X-Y=5$
- (2) Is there a linear equation of the form $aX + bY = c$, where not all a, b, c are zero, for which the points $(1, 2)$, $(2, 3)$ and $(3, 5)$ are solutions.
- (3) Let $P_1 = (a, b) \in \mathbb{R}^2$ and $P_2 = (c, d) \in \mathbb{R}^2$. Find a linear equation for which P_1 and P_2 are solutions. How many equations are there having P_1 and P_2 as solutions?
- (4) Find all the linear equations having $(0, 0) \in \mathbb{R}^2$ as one of its solutions. Justify your answer!
- (5) For distinct numbers $a, b \in \mathbb{R}$, find the linear equation for which $(4, a)$ and $(4, b) \in \mathbb{R}^2$ are solutions.
- (6) Draw the solution set of the equation

$$Y - X = 2.$$

Write down a few solutions of the form $(p, q) \in \mathbb{N}^2$, where $\mathbb{N} = \{1, 2, 3, \dots\}$ and both p and q are prime numbers².

¹If there is only one variable present in the equation then treat the coefficient of the other one to be zero.

²It is not known whether there are infinitely many such pairs! Search on internet "twin prime conjecture".